REQUIREMENT ANALYSIS & USER STORY WRITING

Online Book Store Project

Assignment No. 1

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1. Document Control

The following table captures version history and ownership of this document.

Version	Date	Prepared By	Reviewed By	Status
1.0	04-23-2026	Farooque Sajjad	[Supervisor Name]	Draft

2. Project Overview

The client requires an Online Book Store that gives users the ability to browse, search, and purchase books through a web-based interface. The system must support account management, a structured product catalog, a shopping cart, and secure payment processing. The goal is to deliver a smooth user experience while maintaining data integrity and system reliability.

2.1 Business Objectives

- Allow customers to browse and purchase books without visiting a physical store
- Provide a searchable and filterable book catalog with detailed product information
- Enable users to manage their own accounts, order history, and preferences
- Process payments through a secure and trusted payment gateway
- Support the business in tracking sales, inventory, and user activity

2.2 Scope

This assignment covers requirement analysis, user stories, traceability, and sprint planning for the first release of the Online Book Store. The scope includes all user types: Guest, Registered Customer, and Administrator. Features covered are user authentication, catalog management, search and filtering, cart management, checkout, payment, and basic order tracking. Admin features such as inventory management are included as supporting functionality.

2.3 User Roles

Role	Access Level	Primary Actions
Guest / Visitor	Public	Browse catalog, search books, view book details
Registered Customer	Authenticated	All guest actions plus cart, checkout, payment, orders, wishlist, reviews
Administrator	Admin Panel	Manage catalog, orders, users, and view reports

2.4 Stakeholders

Stakeholder	Role	Responsibility
End Users	Customers	Browse, search, and purchase books
Admin	Store Manager	Manage catalog, orders, and users
QA Engineer	Tester	Validate requirements and test scenarios
Developer	Implementation	Build features based on accepted requirements
Product Owner	Client	Define and prioritize requirements

3. Requirement Analysis

Requirement analysis was conducted using client description, industry standards for e-commerce platforms, and review of similar systems. Requirements are categorized into functional and non-functional groups. Priority is assigned using MoSCoW notation (Must Have, Should Have, Could Have, Won't Have).

3.1 Functional Requirements

The table below lists all functional requirements identified for the system.

ID	Feature	Description	Priority
FR-01	**User Registration**	Users must be able to register with email and password. Email verification is required before login.	Must Have
FR-02	**User Login**	Registered users must be able to log in using email and password. System must support session management.	Must Have
FR-03	**Password Reset**	Users must be able to reset their password via a link sent to their registered email address.	Must Have
FR-04	**Book Catalog**	The system must display a catalog of books with title, author, genre, price, cover image, and availability.	Must Have
FR-05	**Book Search**	The system shall provide a search interface allowing users to query the book catalog based on title, author, or keyword attributes.	Must Have
FR-06	**Filter and Sort**	Users must be able to filter results by genre,	Must Have

		price range, rating, and language. Sorting by price and relevance must be supported.	
FR-07	**Book Details Page**	Each book must have a dedicated detail page with full description, author bio, customer reviews, and stock status.	Must Have
FR-08	**Shopping Cart**	Users must be able to add, update, and remove books from the shopping cart. Cart must persist across sessions for logged-in users.	Must Have
FR-09	**Checkout Process**	The checkout flow must include address entry, order summary, and payment selection before order placement.	Must Have
FR-10	**Payment Integration**	The system shall provide a secure checkout interface that supports multiple payment gateways, including credit/debit cards (Visa, Mastercard) and integrated mobile wallets (e.g., NayaPay, Google Pay).	Must Have
FR-11	**Order Confirmation**	After successful payment, users must receive an order confirmation page and an email with order details.	Must Have
FR-12	**Order History**	Logged-in users must be able to view their past orders with status, items, and invoice details.	Should Have
FR-13	**Wishlist**	Users should be able to save books to a personal wishlist for future reference.	Should Have
FR-14	**Book Reviews**	Authenticated users should be able to submit star ratings and text reviews for books they have purchased.	Should Have
FR-15	**Admin: Catalog Managemet**	Admins must be able to add, edit, and remove books from the catalog, including pricing and stock updates.	Must Have
FR-16	**Admin: Order Management**	Admins must be able to view all orders and update order status (pending, shipped, delivered, cancelled).	Must Have
FR-17	**Admin: User Management**	Admins must be able to view registered users and disable accounts where necessary.	Must Have
FR-18	**Notifications**	The system should send email or SMS notifications for order updates, shipping, and delivery.	Should Have
FR-19	**Discount /**	The system could support discount codes that	Could Have

	Coupons	users apply during checkout.	
FR-20	**Book Recommendations**	The system could display personalized book suggestions based on browsing or purchase history.	Could Have
FR-21	**Mobile App**	A dedicated mobile application for Android and iOS is not planned for this release.	Won't Have
FR-22	**Multi-language Support**	Support for languages other than English is not in scope for this release.	Won't Have
FR-23	**Physical Store Integration**	Integration with a point-of-sale or physical inventory system is out of scope.	Won't Have
FR-24	**Subscription / Membership**	A paid subscription tier with exclusive benefits is not planned for this release.	Won't Have

3.2 Non-Functional Requirements

Non-functional requirements define the quality attributes the system must meet.

ID	Category	Requirement	Acceptance Metric	Priority
NFR-01	**Performance**	Search results must load within an acceptable time under normal load.	Response time under 2 seconds for up to 500 concurrent users	Must Have
NFR-02	**Performance**	The home and catalog pages must load quickly on standard broadband.	Page load under 3 seconds at 80th percentile	Must Have
NFR-03	**Scalability**	The system must handle traffic spikes during promotional periods.	Support 2x normal load without degradation	Must Have
NFR-04	**Security**	All user passwords must be stored using a strong hashing algorithm.	Bcrypt or Argon2 with appropriate work factor	Must Have
NFR-05	**Security**	All data transmission between client and server must be encrypted.	TLS 1.2 or higher enforced on all endpoints	Must Have
NFR-06	**Security**	Payment data must meet industry compliance	PCI-DSS compliance	Must Have

		standards.	required	
NFR-07	**Availability**	The platform must maintain high uptime during peak hours.	99.5% uptime SLA per month	Must Have
NFR-08	**Usability**	The interface must be accessible on common screen sizes.	Fully responsive on desktop, tablet, and mobile	Must Have
NFR-09	**Usability**	Core user flows must be completable without prior training.	Task completion rate above 90% in usability testing	Should Have
NFR-10	**Maintainability**	The codebase must follow documented standards for future updates.	Code coverage above 70% and documented APIs	Should Have
NFR-11	**Reliability**	The system must recover from failures without data loss.	Recovery time under 15 minutes with no data corruption	Must Have
NFR-12	**Compatibility**	The system must work across modern browsers.	Tested on Chrome, Firefox, Safari, and Edge	Must Have

4. User Stories

User stories describe system functionality from the perspective of the end user. Each story follows the standard format: As a [role], I want to [action], so that [benefit].Each story card also shows which FR it addresses to maintain traceability. Acceptance criteria are written in Gherkin syntax and intended to be executed using the Cucumber extension in VSCode.

Note on Gherkin Tool Setup: The Gherkin scenarios in this document were authored and validated using VSCode with the Cucumber (Gherkin) Full Support extension (by Alexander Krechik). Screenshots of the VSCode environment are included in the Appendix.

User roles covered in this section: Guest / Visitor, Registered Customer, and Administrator.

Guest / Visitor Stories

US-01: Browse Book Catalog (Guest)

US-01 - Browse Book Catalog	
As a	guest or visitor
I want to	view a catalog of available books with key details
So that	I can explore what is available before deciding to register or buy
Priority	Must Have
Sprint	Sprint 1
Story Points	5
Linked FRs	FR-04

Acceptance Criteria

```
Feature: Book Catalog

  Scenario: Catalog loads for a guest user
    Given a visitor opens the Online Book Store
    Then the catalog displays a list of available books
    And each book shows title, author, cover image, price, and stock status

  Scenario: Out-of-stock book is marked correctly
    Given a book has zero stock in the system
    When a guest views the catalog
    Then that book shows 'Out of Stock'
    And the Add to Cart button is not visible for that book

  Scenario: Catalog pagination
    Given there are more than 20 books in the system
    When the guest reaches the end of the current page
    Then the system loads the next 20 books or shows pagination controls
```

US-02: Search Books (Guest)

US-02 - Search Books	
As a	guest or visitor
I want to	search books by title, author, or keyword
So that	I can find a specific book quickly without browsing the entire catalog
Priority	Must Have
Sprint	Sprint 1

Story Points	5
Linked FRs	FR-05

Acceptance Criteria

```
Feature: Book Search

  Scenario: Search returns matching results
    Given the visitor is on any page
    When the visitor types 'Atomic Habits' in the search bar and presses Enter
    Then the system displays books matching the query
    And results appear within 2 seconds

  Scenario: No results found
    Given the visitor searches for a term with no matching books
    Then the system shows 'No results found for your search'
    And suggests related categories or popular books

  Scenario: Partial keyword search
    Given the visitor types a partial title or author name
    Then the system returns all books that partially match the input
```

US-03: Filter and Sort (Guest)

US-03 - Filter and Sort	
As a	guest or visitor
I want to	filter books by genre, price, and rating and sort the results
So that	I can narrow down options and find the most relevant book faster
Priority	Must Have
Sprint	Sprint 2
Story Points	5
Linked FRs	FR-06

Acceptance Criteria

```
Feature: Filter and Sort

  Scenario: Filter by genre
    Given the visitor is on the catalog or search results page
    When the visitor selects 'Fiction' from the genre filter
    Then only books tagged as 'Fiction' are displayed

  Scenario: Filter by price range
    Given the visitor sets a price filter of Rs. 500 to Rs. 1500
    Then only books priced within that range are shown
```

```
Scenario: Sort by price low to high
  Given the visitor selects 'Price: Low to High' from the sort dropdown
  Then books reorder with the lowest price appearing first

Scenario: Combined filter and sort
  Given the visitor filters by 'Science' and sorts by 'Rating'
  Then only Science books are shown, ordered by rating descending
```

US-04: View Book Details (Guest)

US-04 - View Book Details	
As a	guest or visitor
I want to	view a dedicated detail page for a specific book
So that	I can read the full description and reviews before deciding to buy
Priority	Must Have
Sprint	Sprint 2
Story Points	3
Linked FRs	FR-07

Acceptance Criteria

```
Feature: Book Detail Page

  Scenario: Book detail page opens
    Given the visitor clicks on a book in the catalog
    Then the system opens the book detail page
    And displays title, author, description, genre, price, cover image, and average rating

  Scenario: Reviews are visible on detail page
    Given the book has at least one customer review
    When the visitor scrolls to the reviews section
    Then reviews appear with reviewer name, star rating, and comment

  Scenario: Add to Cart from detail page (redirects guest)
    Given the visitor is not logged in and clicks 'Add to Cart'
    Then the system redirects to the login page
    And shows 'Please log in to add items to your cart'
```

Registered Customer Stories

US-05: User Registration

US-05 - User Registration	
As a	new visitor
I want to	register an account using my email and password
So that	I can access personalized features like cart and order history
Priority	Must Have
Sprint	Sprint 1
Story Points	5
Linked FRs	FR-01

Acceptance Criteria

```
Feature: User Registration

  Scenario: Successful registration with valid details
    Given the user is on the registration page
    When the user enters a valid email, password, and full name
    And submits the registration form
    Then the system creates the account
    And sends a verification email to the registered address
    And redirects the user to the email confirmation page

  Scenario: Registration with an already registered email
    Given the user is on the registration page
    When the user enters an email that is already in the system
    And submits the registration form
    Then the system displays the error 'This email is already registered'
    And the form is not submitted

  Scenario: Registration with a weak password
    Given the user is on the registration page
    When the user enters a password shorter than 8 characters
    Then the system shows inline validation 'Password must be at least 8 characters'
    And the submit button remains disabled
```

US-06: User Login

US-06 - User Login	
As a	registered user
I want to	log in with my email and password

So that	I can access my account and saved data
Priority	Must Have
Sprint	Sprint 1
Story Points	3
Linked FRs	FR-02

Acceptance Criteria

```
Feature: User Login

  Scenario: Successful login with valid credentials
    Given the user is on the login page
    When the user enters a registered email and correct password
    And clicks the Login button
    Then the system authenticates the user
    And redirects to the home page
    And displays the user's name in the navigation bar

  Scenario: Login with incorrect password
    Given the user is on the login page
    When the user enters a registered email and an incorrect password
    And clicks the Login button
    Then the system displays 'Invalid email or password'
    And the user remains on the login page

  Scenario: Login attempt after 5 failed tries
    Given the user has failed to login 5 consecutive times
    When the user attempts to login again
    Then the system locks the account for 15 minutes
    And displays a message informing the user to try later
```

US-07: Password Reset

US-07 - Password Reset	
As a	registered user
I want to	reset my password via email
So that	I can regain access to my account if I forget my credentials
Priority	Must Have
Sprint	Sprint 1
Story Points	3
Linked FRs	FR-03

Acceptance Criteria

```
Feature: Password Reset

  Scenario: Successful password reset request
    Given the user is on the forgot password page
    When the user enters their registered email
    And clicks Send Reset Link
    Then the system sends a password reset email
    And displays 'Check your inbox for reset instructions'

  Scenario: Password reset with unregistered email
    Given the user is on the forgot password page
    When the user enters an email not in the system
    And clicks Send Reset Link
    Then the system shows the same success message to prevent email enumeration

  Scenario: Expired reset link
    Given the user clicks a password reset link older than 60 minutes
    Then the system displays 'This link has expired. Please request a new one.'
```

US-08: Browse Book Catalog

US-08 - Browse Book Catalog	
As a	visitor or logged-in user
I want to	view a catalog of available books with key details
So that	I can explore what is available before making a purchase decision
Priority	Must Have
Sprint	Sprint 1
Story Points	5
Linked FRs	FR-04

Acceptance Criteria

```
Feature: Book Catalog

  Scenario: Catalog loads on home page
    Given the user navigates to the catalog section
    Then the system displays a list of books
    And each book shows the title, author, cover image, price, and availability status

  Scenario: Catalog displays correct stock status
    Given a book is out of stock in the system
    When the user views that book in the catalog
    Then the catalog entry shows 'Out of Stock'
    And the Add to Cart button is disabled for that book

  Scenario: Catalog pagination
```

```
Given there are more than 20 books in the catalog
When the user reaches the bottom of the page
Then the system loads the next 20 books
Or provides pagination controls to navigate between pages
```

US-09: Search Books

US-09 - Search Books	
As a	visitor or logged-in user
I want to	search books by title, author, or keyword
So that	I can find the specific book I am looking for without browsing the entire catalog
Priority	Must Have
Sprint	Sprint 1
Story Points	5
Linked FRs	FR-05

Acceptance Criteria

```
Feature: Book Search

  Scenario: Search returns matching results
    Given the user is on any page with the search bar visible
    When the user types 'Rich Dad Poor Dab' and presses Enter
    Then the system displays all books matching the query
    And results are shown within 2 seconds

  Scenario: Search with no results
    Given the user is on the search page
    When the user searches for a term that matches no book
    Then the system displays 'No results found for your search'
    And suggests related categories or popular books

  Scenario: Search with partial keyword
    Given the user types a partial title or author name
    Then the system returns all books that partially match the input
```

US-10: Filter and Sort Results

US-10 - Filter and Sort	
As a	visitor or logged-in user

I want to	filter books by genre, price, and rating and sort results
So that	I can narrow down options and find the most relevant book faster
Priority	Must Have
Sprint	Sprint 2
Story Points	5
Linked FRs	FR-06

Acceptance Criteria

```
Feature: Filter and Sort

  Scenario: Filter by genre
    Given the user is on the catalog or search results page
    When the user selects 'Fiction' from the genre filter
    Then only books with the genre 'Fiction' are displayed

  Scenario: Filter by price range
    Given the user applies a price filter of Rs. 500 to Rs. 1500
    Then only books priced between Rs. 500 and Rs. 1500 are shown

  Scenario: Sort by price low to high
    Given the user selects 'Price: Low to High' from the sort dropdown
    Then books are reordered with the lowest price appearing first

  Scenario: Combined filter and sort
    Given the user filters by genre 'Science' and sorts by 'Rating'
    Then the system displays only Science books ordered by rating descending
```

US-11: View Book Details

US-11 - View Book Details	
As a	visitor or logged-in user
I want to	view a detailed page for a specific book
So that	I can read the full description, reviews, and make an informed purchase decision
Priority	Must Have
Sprint	Sprint 2
Story Points	3
Linked FRs	FR-07

Acceptance Criteria

```
Feature: Book Detail Page

  Scenario: User opens a book detail page
    Given the user clicks on a book from the catalog
    Then the system opens the book detail page
    And displays the title, author, description, genre, price, cover image, and average rating

  Scenario: Book detail shows customer reviews
    Given the book has at least one customer review
    When the user scrolls to the reviews section
    Then the reviews are visible with reviewer name, star rating, and comment

  Scenario: Add to cart from detail page
    Given the book is in stock
    When the user clicks 'Add to Cart' on the detail page
    Then the book is added to the cart
    And the cart icon updates to reflect the new item count
```

US-12: Shopping Cart Management

US-12 - Shopping Cart	
As a	logged-in user
I want to	add, update, and remove books from my cart
So that	I can manage my selection before proceeding to checkout
Priority	Must Have
Sprint	Sprint 2
Story Points	8
Linked FRs	FR-08

Acceptance Criteria

```
Feature: Shopping Cart

  Scenario: Add a book to cart
    Given the user is logged in and views an in-stock book
    When the user clicks 'Add to Cart'
    Then the book appears in the cart with quantity set to 1
    And the cart total is updated

  Scenario: Update book quantity in cart
    Given the user has a book in the cart
    When the user changes the quantity to 2
    Then the cart reflects the updated quantity
    And the subtotal for that item is recalculated

  Scenario: Remove a book from cart
```

```
  Given the user has one or more books in the cart
  When the user clicks the remove icon for a specific book
  Then that book is removed from the cart
  And the total is updated accordingly

Scenario: Cart persists after logout and login
  Given the user has items in the cart
  When the user logs out and logs back in
  Then the cart still contains the previously added items
```

US-13: Checkout Process

US-13 - Checkout	
As a	logged-in user
I want to	complete a checkout by providing delivery address and reviewing my order
So that	I can confirm what I am buying and where it should be delivered before payment
Priority	Must Have
Sprint	Sprint 3
Story Points	8
Linked FRs	FR-09

Acceptance Criteria

```
Feature: Checkout

  Scenario: Successful checkout with saved address
    Given the user has items in the cart and is logged in
    When the user proceeds to checkout
    Then the system shows the delivery address section
    And pre-fills any saved address
    And displays an order summary with item names, quantities, and total price

  Scenario: Checkout with missing address
    Given the user proceeds to checkout without a saved address
    When the user skips the address form and clicks Continue
    Then the system highlights the required address fields
    And does not proceed to the payment step

  Scenario: Checkout with empty cart
    Given the user's cart is empty
    When the user navigates to checkout
    Then the system redirects to the cart page
    And displays 'Your cart is empty'
```

US-14: Payment Processing

US-14 - Payment	
As a	logged-in user
I want to	pay for my order using a credit/debit card or mobile wallet
So that	I can complete my purchase securely without leaving the platform
Priority	Must Have
Sprint	Sprint 3
Story Points	8
Linked FRs	FR-10

Acceptance Criteria

```
Feature: Payment

  Scenario: Successful payment with card
    Given the user is on the payment page
    When the user enters valid card details and clicks Pay Now
    Then the system processes the payment via the payment gateway
    And redirects the user to the order confirmation page

  Scenario: Payment fails due to insufficient funds
    Given the user is on the payment page
    When the user submits a card with insufficient funds
    Then the system displays 'Payment failed. Please try a different payment method.'
    And the order is not placed

  Scenario: Payment using mobile wallet
    Given the user selects a mobile wallet option
    When the user completes the wallet authentication
    Then the payment is processed
    And an order confirmation is generated
```

US-15: Order Confirmation and Email

US-15 - Order Confirmation	
As a	logged-in user
I want to	receive an order confirmation on screen and via email
So that	I have a record of my purchase and know the order was placed successfully

Priority	Must Have
Sprint	Sprint 3
Story Points	3
Linked FRs	FR-11

Acceptance Criteria

```
Feature: Order Confirmation

  Scenario: Confirmation page after payment
    Given the user has completed payment
    Then the system displays a confirmation page
    And shows the order ID, list of purchased books, total amount, and estimated delivery

  Scenario: Confirmation email is received
    Given the order is placed successfully
    Then the system sends a confirmation email to the user's registered address
    And the email includes the order ID, items, and delivery details
```

US-16: Order History

US-16 - Order History	
As a	logged-in user
I want to	view all my past orders with their status and details
So that	I can track what I have purchased and refer back to invoices
Priority	Should Have
Sprint	Sprint 3
Story Points	5
Linked FRs	FR-12

Acceptance Criteria

```
Feature: Order History

  Scenario: View order history
    Given the user is logged in
    When the user navigates to My Orders
    Then the system displays all past orders
    And each entry shows the order ID, date, status, and total amount

  Scenario: View order details
    Given the user is on the order history page
```

```
When the user clicks on a specific order
Then the system opens a detail view
And shows each book in the order with its price, quantity, and shipping info
```

US-17: Wishlist

US-17 - Wishlist	
As a	logged-in user
I want to	save books to a wishlist for later consideration
So that	I can keep track of books I am interested in without adding them to cart
Priority	Should Have
Sprint	Sprint 4
Story Points	3
Linked FRs	FR-13

Acceptance Criteria

```
Feature: Wishlist

  Scenario: Add book to wishlist
    Given the user is logged in and viewing a book
    When the user clicks the wishlist icon
    Then the book is added to the user's wishlist
    And the icon changes to indicate it is saved

  Scenario: Remove book from wishlist
    Given the user has books in the wishlist
    When the user clicks the wishlist icon on a saved book
    Then the book is removed from the wishlist

  Scenario: Move book from wishlist to cart
    Given the user is viewing the wishlist
    When the user clicks 'Add to Cart' on a wishlist item
    Then the book is added to the cart
    And optionally removed from the wishlist
```

US-18: Book Reviews

US-18 - Book Reviews

As a	logged-in user who purchased the book
I want to	submit a star rating and written review for a book
So that	other users can make informed purchase decisions based on honest feedback
Priority	Should Have
Sprint	Sprint 4
Story Points	5
Linked FRs	FR-14

Acceptance Criteria

```
Feature: Book Reviews

  Scenario: Submit a review for a purchased book
    Given the user has purchased the book
    When the user navigates to the book page and submits a 4-star rating with a comment
    Then the review is saved and displayed on the book detail page
    And the book's average rating is updated

  Scenario: Attempt to review without purchase
    Given the user is logged in but has not purchased the book
    When the user attempts to submit a review
    Then the system displays 'You must purchase this book to leave a review'

  Scenario: Review moderation
    Given a review contains flagged content
    When the moderation check runs
    Then the review is held for admin approval before being published
```

US-19: Notifications

US-19 - Order Notifications	
As a	registered customer
I want to	receive email or SMS updates on my order status
So that	I stay informed about shipping and delivery without checking the app
Priority	Should Have
Sprint	Sprint 3
Story Points	3
Linked FRs	FR-18

Acceptance Criteria

```
Feature: Order Notifications

  Scenario: Notification on order placement
    Given the user places an order
    Then the system sends an email confirming the order

  Scenario: Notification when order is shipped
    Given an admin updates the order status to 'Shipped'
    Then the system sends a shipping notification to the customer
    And the notification includes the tracking reference

  Scenario: Notification on delivery
    Given the order status is updated to 'Delivered'
    Then the system sends a delivery confirmation to the customer
```

US-20: Discount Codes

US-20 - Discount / Coupons	
As a	registered customer
I want to	apply a discount code at checkout
So that	I can take advantage of promotions and pay a reduced price
Priority	Could Have
Sprint	Sprint 4
Story Points	5
Linked FRs	FR-19

Acceptance Criteria

```
Feature: Discount Codes

  Scenario: Valid discount code applied
    Given the user is on the checkout page
    When the user enters a valid discount code and clicks Apply
    Then the discount is deducted from the order total
    And the updated total is shown before payment

  Scenario: Invalid or expired code
    Given the user enters an invalid or expired code
    Then the system shows 'This code is invalid or has expired'
    And the original total remains unchanged

  Scenario: One code per order
```

```
Given the user has already applied one code
When the user tries to apply a second code
Then the system shows 'Only one discount code can be applied per order'
```

US-21: Book Recommendations

US-21 - Book Recommendations	
As a	registered customer
I want to	see personalized book suggestions based on my activity
So that	I can discover new books relevant to my interests easily
Priority	Could Have
Sprint	Sprint 4
Story Points	5
Linked FRs	FR-20

Acceptance Criteria

```
Feature: Book Recommendations

  Scenario: Recommendations shown on home page
    Given the user has browsed or purchased books previously
    When the user visits the home page
    Then a 'Recommended for You' section appears
    And it shows books related to the user's history

  Scenario: No history available
    Given the user has no browsing or purchase history
    When the user visits the home page
    Then the system shows 'Popular Books' or 'New Arrivals' as a default
```

Administrator Stories

US-22: Admin - Catalog Management

US-22 - Admin Catalog Management	
As a	store administrator
I want to	add, edit, and remove books from the catalog
So that	the product listing stays accurate and up to date for customers

Priority	Must Have
Sprint	Sprint 2
Story Points	8
Linked FRs	FR-15

Acceptance Criteria

```
Feature: Admin Catalog Management

  Scenario: Add a new book
    Given the admin is logged in and on the catalog management page
    When the admin fills in the book title, author, genre, price, stock, and cover image
    And clicks Save
    Then the new book appears in the catalog
    And is visible to users on the storefront

  Scenario: Edit an existing book
    Given the admin opens an existing book's edit form
    When the admin updates the price and clicks Save
    Then the updated price is reflected on the book detail page

  Scenario: Remove a book
    Given the admin selects a book from the catalog
    When the admin clicks Delete and confirms the prompt
    Then the book is removed from the catalog
    And no longer visible to users
```

US-23: Admin - Order Management

US-23 - Admin Order Management	
As a	store administrator
I want to	view all orders and update their status
So that	I can process, dispatch, and close orders in a timely manner
Priority	Must Have
Sprint	Sprint 3
Story Points	5
Linked FRs	FR-16

Acceptance Criteria

```
Feature: Admin Order Management
```

```
Scenario: View all orders
  Given the admin is on the order management page
  Then the system lists all orders with ID, customer name, date, total, and current status

Scenario: Update order status
  Given the admin opens an order
  When the admin changes the status from 'Pending' to 'Shipped'
  And saves the change
  Then the order status updates
  And the customer receives a notification email

Scenario: Filter orders by status
  Given the admin is on the order management page
  When the admin selects 'Pending' from the status filter
  Then only pending orders are displayed
```

US-24: Admin - User Management

US-24 - Admin User Management	
As a	store administrator
I want to	view registered users and manage their accounts
So that	I can maintain platform safety and handle account issues
Priority	Must Have
Sprint	Sprint 4
Story Points	5
Linked FRs	FR-17

Acceptance Criteria

```
Feature: Admin User Management

  Scenario: View all users
    Given the admin is on the user management page
    Then a list of all registered users is displayed
    And each entry shows name, email, registration date, and account status

  Scenario: Disable a user account
    Given the admin selects a user and clicks Disable Account
    When the admin confirms the action
    Then the account status changes to 'Disabled'
    And the user cannot log in until the account is re-enabled

  Scenario: Re-enable a disabled account
    Given the admin selects a disabled account
    When the admin clicks Enable Account
    Then the account status changes to 'Active'
    And the user can log in again
```

```
Scenario: Search users by email
  Given the admin types a partial email in the search box
  Then the user list filters to show matching accounts only
```

5. Requirements Traceability Matrix

The traceability matrix links every functional requirement to the user story that covers it. This ensures that no requirement has been missed and each user story has a clear purpose tied back to a stated need. Green rows indicate the requirement is fully covered by at least one user story.

FR ID	FR Title	US ID	User Story Title	Priority	Status
FR-01	User Registration	US-05	User Registration	Must Have	Covered
FR-02	User Login	US-06	User Login	Must Have	Covered
FR-03	Password Reset	US-07	Password Reset	Must Have	Covered
FR-04	Book Catalog	US-01,US-08	Browse Book Catalog	Must Have	Covered
FR-05	Book Search	US-02,US-09	Search Books	Must Have	Covered
FR-06	Filter and Sort	US-03,US-10	Filter and Sort	Must Have	Covered
FR-07	Book Details Page	US-04,US-11	View Book Details	Must Have	Covered
FR-08	Shopping Cart	US-12	Shopping Cart	Must Have	Covered
FR-09	Checkout Process	US-13	Checkout Process	Must Have	Covered
FR-10	Payment Integration	US-14	Payment Processing	Must Have	Covered
FR-11	Order Confirmation	US-15	Order Confirmation	Must Have	Covered
FR-12	Order History	US-16	Order History	Should	Covered

				Have	
FR-13	Wishlist	US-17	Wishlist	Should Have	Covered
FR-14	Book Reviews	US-18	Book Reviews	Should Have	Covered
FR-15	Admin: Catalog Mgmt	US-22	Admin Catalog Management	Must Have	Covered
FR-16	Admin: Order Mgmt	US-23	Admin Order Management	Must Have	Covered
FR-17	Admin: User Mgmt	US-24	Admin User Management	Must Have	Covered
FR-18	Notifications	US-19	Order Notifications	Should Have	Covered
FR-19	Discount / Coupons	US-20	Discount Codes	Could Have	Covered
FR-20	Book Recommendations	US-21	Book Recommendations	Could Have	Covered
FR-21	Mobile App	N/A	Out of Scope - Won't Have	Won't Have	Not Covered
FR-22	Multi-language Support	N/A	Out of Scope - Won't Have	Won't Have	Not Covered
FR-23	Physical Store Integration	N/A	Out of Scope - Won't Have	Won't Have	Not Covered
FR-24	Subscription / Membership	N/A	Out of Scope - Won't Have	Won't Have	Not Covered

6. Sprint Planning

Stories are organized across four two-week sprints based on priority and dependency. Story points use the Fibonacci scale. All Must Have stories are completed by Sprint 3. Should Have and Could Have stories are handled in Sprint 4.

6.1 Sprint Summary

Sprint	Focus Area	Stories	Points

Sprint 1	Authentication and Catalog Foundation	US-01,02,05,06,07,08,09	31
Sprint 2	Filtering, Cart, Book Detail, Admin Catalog	US-03,04,10,11,12,22	32
Sprint 3	Checkout, Payment, Orders, Admin Orders, Notifications	US-13,14,15,16,19,23	32
Sprint 4	Wishlist, Reviews, Discounts, Recommendations, Admin Users	US-17,18,20,21,24	23

6.2 Detailed Sprint Backlog

Story ID	Title	Points	Sprint
US-01	Browse Book Catalog (Guest)	5	Sprint 1
US-02	Search Books (Guest)	5	Sprint 1
US-05	User Registration	5	Sprint 1
US-06	User Login	3	Sprint 1
US-07	Password Reset	3	Sprint 1
US-08	Browse Book Catalog (User)	5	Sprint 1
US-09	Search Books (User)	5	Sprint 1
US-03	Filter and Sort	5	Sprint 2
US-04	View Book Details	3	Sprint 2
US-10	Filter and Sort (User)	5	Sprint 2
US-11	View Book Details (User)	3	Sprint 2
US-12	Shopping Cart Management	8	Sprint 2
US-22	Admin: Catalog Management	8	Sprint 2
US-13	Checkout Process	8	Sprint 3
US-14	Payment Processing	8	Sprint 3
US-15	Order Confirmation	3	Sprint 3
US-16	Order History	5	Sprint 3
US-19	Order Notifications	3	Sprint 3
US-23	Admin: Order Management	5	Sprint 3

US-17	Wishlist	3	Sprint 4
US-18	Book Reviews	5	Sprint 4
US-20	Discount Codes	5	Sprint 4
US-21	Book Recommendations	5	Sprint 4
US-24	Admin: User Management	5	Sprint 4
TOTAL	**Sum of all Story Points**	**118**	

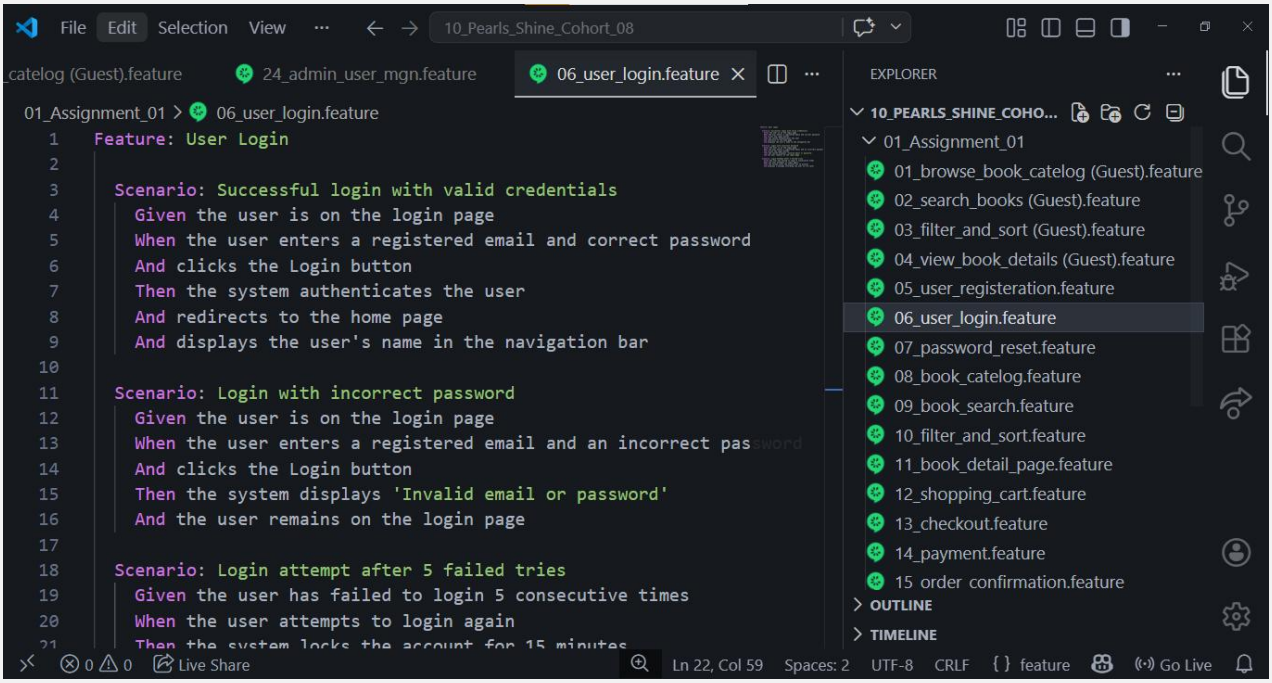
7. Appendix

7.1 Gherkin Tool Setup (VSCode)

All Gherkin scenarios in this document were written and reviewed using Visual Studio Code (VSCode) with the Cucumber (Gherkin) Full Support extension. Follow these steps to set up the same environment:

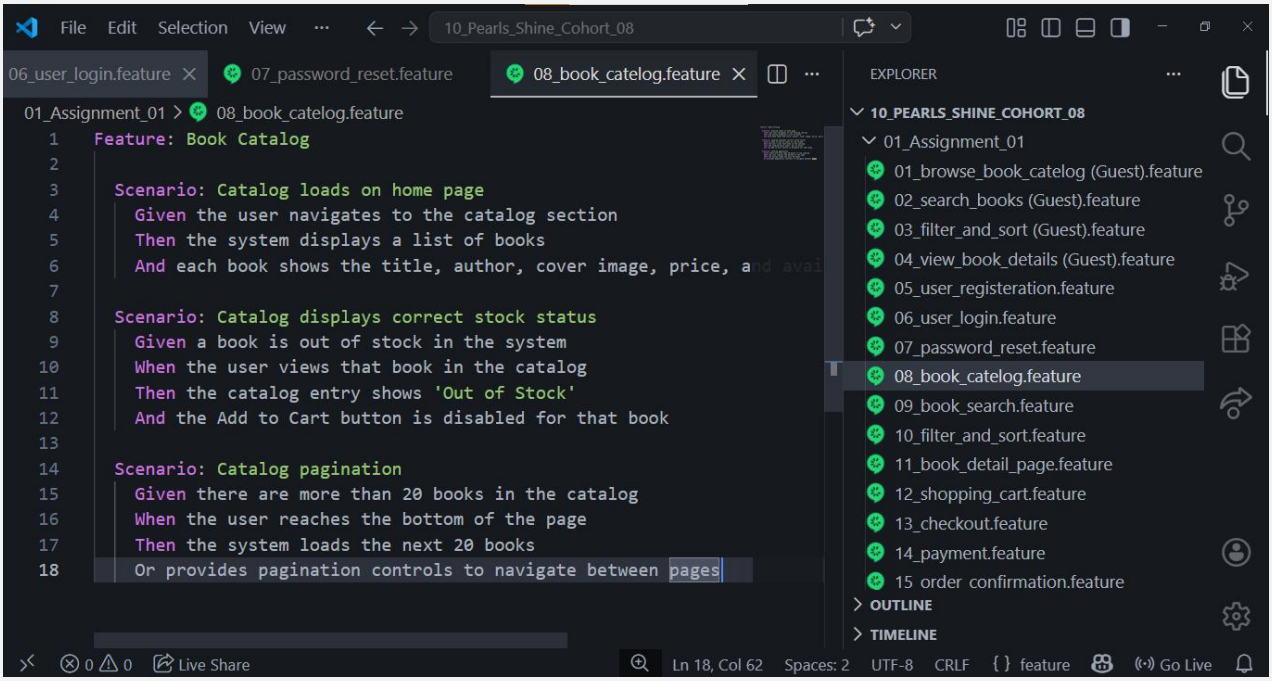
1. Download and install Visual Studio Code from https://code.visualstudio.com
2. Open the Extensions panel (Ctrl+Shift+X) and search for 'Cucumber (Gherkin) Full Support'
3. Install the extension by Alexander Krechik
4. Create a file with the .feature extension (e.g., 01_user_login.feature)
5. Start typing Gherkin keywords such as Feature, Scenario, Given, When, Then to get syntax highlighting and auto-complete

The following screenshots show the .feature files authored in VSCode with syntax highlighting enabled.



06_user_login.feature

Figure 1: Gherkin syntax for Login feature in VSCode with Cucumber extension



08_book_catelog.feature

Figure 2: Gherkin syntax for Registration feature in VSCode

7.2 Glossary

Term	Definition
User Story	A short description of a feature told from the user's perspective
Gherkin	A plain-text language used to describe test scenarios in Given-When-Then format
Acceptance Criteria	Conditions that a feature must meet to be accepted by the product owner
MoSCoW	Prioritization method: Must Have, Should Have, Could Have, Won't Have
Story Points	A relative unit of effort estimation used in agile development
Sprint	A fixed time period (usually 2 weeks) for completing a set of user stories
FR	Functional Requirement - what the system must do
NFR	Non-Functional Requirement - how well the system must perform
PCI-DSS	Payment Card Industry Data Security Standard for secure payment processing
Traceability Matrix	A table linking requirements to user stories to ensure complete coverage